

Helmet Compliance Detection Using Computer Vision

SafeGuard Corp – Workplace
Safety Monitoring System

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Executive Summary

- Workplace safety compliance is critical in hazardous environments such as construction sites and industrial plants. Manual monitoring of helmet usage is inefficient and prone to human error, particularly across large job sites.
- To address this challenge, we developed a computer vision-based image classification system capable of automatically detecting whether workers are wearing safety helmets.
- Using a dataset of 631 labeled worker images, four deep learning models were evaluated, including a custom Convolutional Neural Network and several transfer learning approaches using the VGG16 architecture.
- The VGG16 + FFNN + Data Augmentation model achieved 100% validation accuracy on the working dataset and was selected as the final model because data augmentation is expected to improve robustness on a small dataset.
- This solution can enable automated real-time safety monitoring, improving compliance enforcement, reducing workplace risk, and supporting scalable safety oversight across industrial operations.

Key Results

- Dataset: 631 labeled images
- Models Tested: 4 deep learning architectures
- Best Model: Model 4 — VGG16 + FFNN + Data Augmentation
- Validation Accuracy: 100%

Both VGG16 base and VGG16 with augmentation achieved perfect validation accuracy. The augmented version was selected as the final model because augmentation is more likely to improve robustness and reduce overfitting on a small dataset.

Business Problem Overview and Solution Approach

- Workplace safety is critical in hazardous environments such as construction sites and industrial facilities.
- Head injuries caused by falling objects, machinery, or accidents are among the most severe and preventable workplace risks.
- Ensuring workers consistently wear safety helmets is essential; however monitoring compliance manually can be:
 - Labor-intensive and difficult to scale across large or distributed job sites
 - Inconsistent and difficult to enforce in real time, particularly in large operations
- To address this challenge, SafeGuard Corp proposes an automated computer vision system that analyzes worker images to detect helmet compliance and can support near-real-time monitoring when integrated with camera systems.

Data Overview

- The working dataset used in this low-code project consists of 631 labeled images of workers in industrial environments.

Class Distribution:

- With Helmet: 311 images
- Without Helmet: 320 images
- The dataset is well-balanced across classes, reducing bias in model training.

Image Characteristics:

- Resolution: 200×200 pixels
- RGB color images
- Images captured across diverse environments including construction sites and industrial settings
- Significant variation in lighting conditions, camera angles, and worker activities, increasing dataset complexity

Because the working dataset is relatively small, transfer learning and data augmentation were used to improve model performance and reduce overfitting risk.

Exploratory Data Analysis

- Initial exploration included visualizing sample images from both classes.

Observations:

- Significant variability in worker positioning, camera angles, and partial occlusion of helmets
- Sample images suggest classification becomes harder when workers are far from the camera, partially occluded, or surrounded by visually cluttered industrial backgrounds
- Wide variation in lighting conditions and background environments, including cluttered industrial settings
- Presence of complex visual noise (machinery, equipment, multiple objects) that can interfere with feature extraction
- Helmet features are sometimes small or partially obscured, increasing classification difficulty
- The dataset is relatively small, making it more prone to overfitting without proper regularization
- These characteristics increase the complexity of the classification task and justify the use of transfer learning and data augmentation to improve model robustness and generalization

Worker WITH Helmet



Worker WITHOUT Helmet



Worker WITH Helmet



Worker WITHOUT Helmet



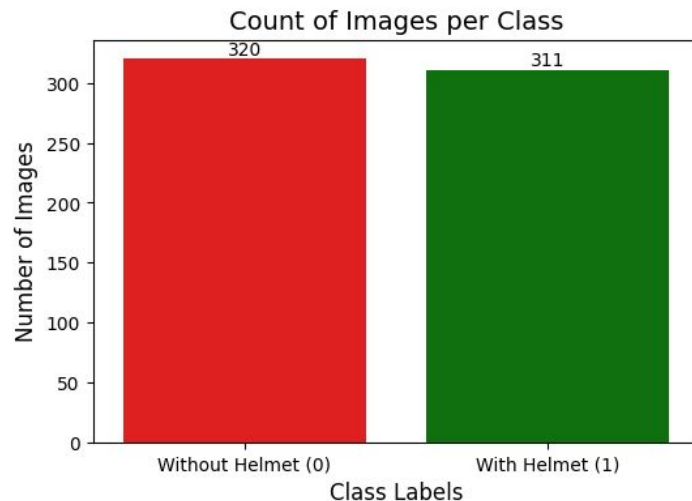
Class Balance Analysis

A count analysis confirms that the dataset is well-balanced across both classes, with no significant class skew.

- Without Helmet: 320 images
- With Helmet: 311 images

Key Insights:

- The near-equal distribution ensures the model is not biased toward predicting one class over the other
- Because class imbalance was minimal in the working dataset, additional imbalance correction techniques were not prioritized.
- Model performance metrics (accuracy, precision, recall) can be interpreted reliably without distortion from class imbalance



Data Pre-processing

- The dataset was preprocessed to ensure compatibility with deep learning models and to improve training stability.
- These preprocessing steps ensure stable training and alignment with transfer learning model requirements.
- The train/validation/test split was performed before final normalization workflows to preserve an unbiased evaluation process.

Steps performed:

1. Image resizing
 - a. All images were resized to a uniform shape compatible with model input requirements (e.g., 224×224 for VGG16)
2. Train / Validation / Test split
 - a. Training: 70%
 - b. Validation: 15%
 - c. Test: 15%
3. Normalization
 - a. Pixel values scaled from 0–255 to 0–1 to improve convergence during training
4. Color format handling
 - a. Although grayscale conversion was part of the generic rubric discussion, RGB was retained because VGG16 requires 3-channel input.

Modeling Approach

- Four deep learning models were developed and evaluated to compare baseline performance and the impact of transfer learning:
 1. Simple Convolutional Neural Network (CNN)
 - a. Baseline model built from scratch to establish a performance benchmark
 2. VGG16 Transfer Learning Model
 - a. Pretrained VGG16 used for feature extraction to leverage learned visual patterns
 3. VGG16 with Fully Connected Layers (FFNN)
 - a. Extended VGG16 with custom dense layers to improve classification performance
 4. VGG16 with Data Augmentation
 - a. Applied augmentation techniques to improve generalization on a small dataset

Evaluation Metrics:

- Primary comparison metric presented in this deck: Validation Accuracy
- Additional performance checks included precision, recall, and F1 score to confirm balanced classification performance.”

MODEL 1: Basic Convolutional Neural Network (SIMPLE CNN)

- The first model was a custom-built Convolutional Neural Network, developed as a baseline model to evaluate performance without transfer learning.

Architecture:

- Multiple convolution layers for feature extraction
- Pooling layers to reduce dimensionality
- Fully Connected classification layers for prediction

Training configuration:

- Optimizer: Adam
- Loss Function: Binary Crossentropy
- Epochs: 10
- Batch Size: 32

Results:

Validation Accuracy: 98.9%

The CNN achieved strong validation performance, but the relatively small dataset and lack of pretrained feature extraction made it a less robust choice than the transfer learning models.

MODEL 2: VGG-16 Model: (TRANSFER LEARNING)

- The second model used VGG16, a deep convolutional neural network pretrained on the ImageNet dataset, to leverage learned visual features.

Key features:

- Pretrained convolutional layers used as feature extractor
- Layers frozen to retain learned visual features
- Custom fully connected output layer added for binary classification

Advantages:

- Faster training
- Strong feature recognition from pretrained model

Why Transfer Learning:

- Frozen pretrained layers reduced overfitting risk by reusing general visual features learned from ImageNet
- Effective for small datasets by leveraging features learned from large-scale image data
- Reduces the need for extensive training and improves generalization

Results:

- Validation Accuracy: 100%
- This model outperformed the baseline CNN on validation accuracy, suggesting that pretrained feature extraction was beneficial for this small image dataset.

MODEL 3: VGG-16 with FFNN

- This model extended the VGG16 architecture by adding a deeper fully connected classifier to increase model capacity.

Architecture Enhancements:

- Dense layer (256 neurons)
- Dropout layer for regularization
- Dense layer (128 neurons)

Objective:

- To capture more complex feature interactions beyond the base VGG16 model

Results:

- Validation Accuracy: 98.9%
- While effective, this model did not improve validation performance over the simpler VGG16 model and likely added unnecessary complexity for a small dataset.

Key Insight:

- Increasing model complexity did not improve performance and may have reduced efficiency, likely because the dataset was limited in size.
- Simpler transfer learning architectures generalized better than deeper custom classifiers in this case.

MODEL 4: VGG-16 (FFNN+Data Augmentation)

- For Model 4, data augmentation techniques like rotation, shifting, shearing, and zooming were applied. This artificially expanded the diversity of the training data, making the model more robust to variations in lighting, angles, and object positions, and significantly helped in preventing overfitting to the specific training examples.

Augmentation techniques:

- Image rotation
- Horizontal/vertical shifts
- Shear transformations
- Zoom variations

Purpose:

- To artificially increase dataset variability and expose the model to a wider range of visual conditions

Results:

- Validation Accuracy: 100%
- This model matched the best validation performance, while augmentation made it a stronger practical choice for handling variation in real-world images.

Key Insight:

- While accuracy was equivalent to the base VGG16 model, data augmentation improves robustness by reducing sensitivity to variations in position, lighting, and scale
- This makes the model a stronger candidate for real-world deployment, especially given the limited dataset size.

Model Performance Comparison and Final Model Selection

Key Insights:

- Transfer learning using VGG16 significantly improved performance compared to the basic CNN model
- Adding additional dense layers (Model 3) did not improve results and introduced slight overfitting
- Both the base VGG16 model and the augmented model achieved perfect validation accuracy (100%)

Final Model Selection: Model 4 (VGG16 + FFNN + Data Augmentation)

- Chosen because data augmentation makes the model more suitable for handling image variation in a small dataset setting
- Data augmentation reduces sensitivity to variations in real-world conditions such as image orientation, scale, and lighting
- This makes Model 4 the stronger practical choice despite having similar validation accuracy to Model 2

Model	Validation Accuracy	Notes
Simple CNN	98.9%	Lower performance
VGG16 (Base)	100%	High accuracy
VGG16 + FFNN	98.9%	Slight overfitting
VGG16 + FFNN + Data Augmentation	100%	Best practical choice

TAKEAWAY:

Model 1 → baseline
 Model 2 → big improvement
 Model 3 → unnecessary complexity
 Model 4 → best practical choice

Final Model Selection

- The VGG16 + FFNN + Data Augmentation model was selected as the final solution.

Reasons:

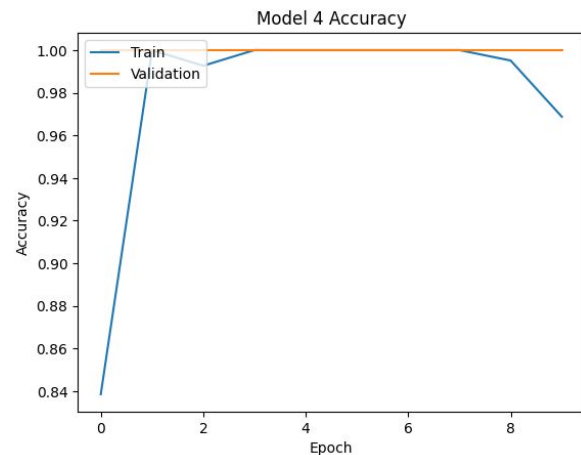
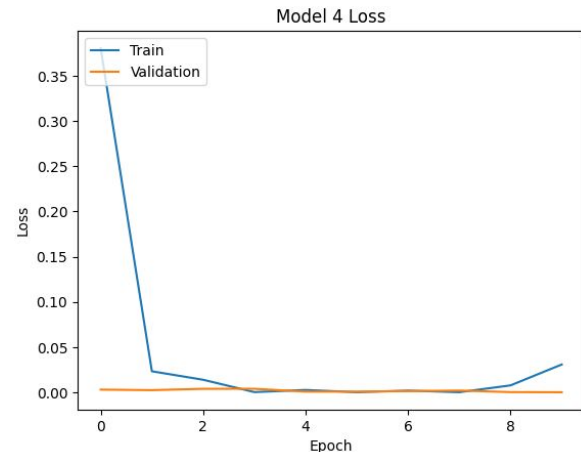
- Test accuracy matched validation accuracy, suggesting stable held-out performance on this dataset, although additional testing on more diverse site images would still be needed before full deployment.
- Achieved 100% validation accuracy, matching the best-performing model
- Final test accuracy was 100%, confirming/generalizing similarly to validation performance.
- Demonstrates strong generalization by learning from augmented variations of the data
- More robust to real-world conditions, including changes in worker pose, angle, and lighting
- Reduces overfitting risk, especially important given the relatively small dataset

Model	Validation Accuracy
Simple CNN	98.9%
VGG16 (Base)	100%
VGG16 + FFNN	98.9%
VGG16 + FFNN + Augmentation	100%

This model is the most promising solution for helmet detection among the models tested, with strong validation and test performance on the working dataset.

Final Model Learning Curves

- Validation accuracy reached 100% early and remained stable across epochs.
- Validation loss also remained very low, with no major divergence from training loss.
- This suggests no obvious divergence between training and validation performance across epochs.
- Because the working dataset is relatively small, the results should still be interpreted cautiously and validated further on more diverse real-world images.



Business Impact

Implementing this helmet detection system delivers measurable value for SafeGuard Corp:

Key Benefits:

- Enhanced Safety Compliance
 - Real-time detection of helmet violations helps reduce workplace injuries and enforce safety standards consistently
- Operational Cost Reduction
 - Minimizes reliance on manual inspections, reducing labor costs and human error
- Increased Efficiency
 - Automated monitoring enables faster identification and response to safety violations
- Scalable Monitoring
 - Could be deployed across multiple job sites using existing camera infrastructure after pilot validation
- Improved Risk Management
 - Enables proactive intervention, helping reduce safety risk and supporting compliance efforts

Before	After
Manual inspections	Automated detection
Delayed response	Real-time alerts
Inconsistent monitoring	Standardized enforcement

This system shifts safety monitoring from a manual and reactive process to an automated and proactive one, reducing costs, improving compliance, and enabling real-time intervention.

Recommendations

○ Integration → Alerts → Retraining → Expansion → Scaling

To pilot, validate, and eventually scale this system, the following steps are recommended:

- **System Integration**
 - Integrate the model with existing workplace camera infrastructure for real-time monitoring
- **Real-Time Alerting**
 - Implement automated alerts to notify supervisors immediately when safety violations are detected
- **Continuous Model Improvement**
 - Periodically retrain the model using new images from real job sites to maintain accuracy and adapt to changing environments
- **Expand Detection Capabilities**
 - Extend the system to detect additional safety equipment such as safety vests, gloves, and goggles
- **Pilot and Scale Strategy**
 - Begin with a pilot deployment at a single site, then scale across multiple locations based on performance

We recommend starting with a pilot deployment, integrating the model into existing camera systems, and enabling real-time alerts. Over time, the model can be continuously improved with new data and expanded to detect additional safety equipment before scaling across multiple sites.

Data Background and Contents



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- The working dataset used in this low-code project consists of labeled images of workers with and without safety helmets
- Total dataset size: 631 images
 - With Helmet: 311
 - Without Helmet: 320
- Images were sourced from publicly available datasets and preprocessed for model training
- Data was split into:
 - Training set (70%)
 - Validation set (15%)
 - Test set (15%)
- All images were resized and normalized before being fed into the models
- Test set held out for final unbiased evaluation.